

Performance Testing Report

Date	Team ID	Project Name	Max Marks
07 July 2026	*****	StudyAI – AI Powered Smart Study Companion	5 Marks

Step 1: Test Scenarios & Tooling

S.No	Test Type	Scenario / API Endpoint Tested	Tool(s) Used
1	Load Testing	Simulating 10 concurrent users hitting /upload and /summary	Locust
2	Stress Testing	Increasing concurrent users on /summary until Groq API rate limits or errors appear	Locust
3	Spike Testing	Sudden burst of users calling /upload simultaneously after idle period	Locust
4	Endurance Testing	Sustained moderate load on /upload over an extended run to check for memory leaks/file handle issues	Locust

Step 2: Key Performance Indicators (KPIs) & Results (⚠ placeholder — not yet measured)

S.No	Metric (KPI)	Target Threshold	Actual Result	Status
1	Average Response Time (/upload)	< 300 ms	[Pending real test]	[Pending]
2	Average Response Time (/summary)	< 3000 ms (bound by Groq API latency, not local processing)	[Pending real test]	[Pending]
3	Throughput (Requests/sec)	> 5 RPS (single-instance Flask dev server)	[Pending real test]	[Pending]
4	Error Rate	< 5% (allowing for Groq rate-limit throttling under burst load)	[Pending real test]	[Pending]
5	CPU/Memory Utilization	< 80% under load	[Pending real test]	[Pending]